Session 2: Hands-on - LSTM Drought Monitoring Lab

Mindanao Case Study with GEE-Sourced NDVI Forecasting

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Session Agenda

- **Duration:** 2.5 hours (150 minutes)
- **Flow:**
  - 0-15: Kickoff & objectives
  - 15-45: Module 1 - Connect to GEE & load NDVI (live or cached)
  - 45-75: Module 2 - Tidy data & build sequences
  - 75-115: Module 3 - Configure PyTorch LSTM workflow
  - 115-135: Module 4 - Train & monitor
  - 135-145: Module 5 - Evaluate & drought decisions
  - 145-150: Reflection & wrap-up

Learning Objectives

- Retrieve Sentinel-2 NDVI time series for Mindanao provinces using Google Earth Engine
- Transform GEE exports into supervised sliding-window datasets (NDVI-only baseline)
- Train, tune, and monitor a PyTorch LSTM with strict temporal validation
- Compare forecasts with persistence and seasonal baselines (MAE/RMSE)
- Translate forecasts into drought alerts and score them (precision/recall)

Resources

- Student notebook: `day4/notebooks/day4_session2_lstm_drought_lab_STUDENT.ipynb`
- Instructor notebook: `day4/notebooks/day4_session2_lstm_drought_lab_INSTRUCTOR.ipynb`
- Default data: cached monthly NDVI sample `day4/data/mindanao_ndvi_sample.csv` (2018–2023, two provinces), so the lab runs without Earth Engine
- Optional live data: Earth Engine Python API + a Cloud project registered for Earth Engine: `ee.Initialize(project="your-cloud-project")`
- Reference: PyTorch `nn.LSTM` documentation

Module 0: Setup & Context

Focus

- Ground the Mindanao drought challenge and toolkit
- Ensure Earth Engine + PyTorch environment is ready

Actions

- Quick NDVI refresher tied to agriculture risk
- Open the student notebook in Colab (CPU runtime is enough)
- Optional: set `USE_EARTH_ENGINE = True` and your Cloud project ID; authenticate once per user
- Why PyTorch here? Session 1 used Keras: same LSTM, second framework

Module 1: Authenticate & Fetch NDVI

Focus

- Pull monthly Sentinel-2 NDVI for Bukidnon & South Cotabato straight from GEE

Actions

- Define province areas (rough bounding boxes; use official boundaries for real work)
- GEE pipeline: S2 SR harmonised → SCL cloud/shadow mask → NDVI → monthly median → ESA WorldCover cropland mask → province mean
- One request per province and year; save to CSV; fall back to the cached sample if anything fails

Module 2: Tidy & Build Sequences

Focus

- Prepare NDVI-only inputs for sequence modelling

Actions

- Inspect gaps/outliers; impute or drop empty months
- Create 12-month lookback / 1-month horizon windows per province
- Temporal split: train on early years, validate on the next, test on the last year
- Min-max scale using the **training period only** (no leakage)

Module 3: Configure PyTorch LSTM

Focus

- Assemble the modelling pipeline around real NDVI data

Actions

- Wrap sequences in custom `Dataset` / `DataLoader`
- Define 2-layer LSTM (64 → 32 units, dropout 0.2) + dense head (same design as the Keras model in Session 1)
- Select optimizer (Adam 1e-3), loss (MSE), metrics (MAE)

Module 4: Train & Monitor

Focus

- Fit the model while checking for convergence and overfitting

Actions

- Train up to 150 epochs with early stopping on validation loss (patience 20); keep the best weights
- Track train/val MSE + MAE; adjust learning rate/batch if unstable
- Fixed random seeds for reproducibility; CPU training takes under a minute

Module 5: Evaluate & Drive Decisions

Focus

- Convert forecasts into drought intelligence for stakeholders

Actions

- Invert scaling and plot actual vs predicted NDVI per province
- MAE/RMSE vs **persistence** (next = this month) and **seasonal naive** (next = same month last year) baselines
- NDVI < 0.40 alert table with precision/recall (an illustrative threshold, not an official criterion)

Reflection & Wrap-up

- Key lessons: accessing satellite archives, disciplined validation, explainable alerts
- Capture open questions for Session 3 (XAI & emerging AI)
- Stretch: add rainfall / ONI / soil moisture as inputs, forecast NDVI anomalies, predict 3 months ahead, explain with Integrated Gradients